Vipul Sapra

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EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Identity Fraud Analytics, Anomaly Detection, Behavioral Biometrics, Predictive Modeling, Account Takeover (ATO) Mitigation, Dimensional Data Modeling, Real-Time Data Processing, Identity Lifecycle Management, Fraud Risk Scoring, Data Governance & Quality Audits

Programming Languages & Tools: Python, SQL, PySpark, Azure Data Factory, Azure Synapse, Databricks, AWS (S3, Redshift), Snowflake, Power BI, Tableau, Scikit-Learn, XGBoost, Airflow, KIRO IDE

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Senior Delivery Consultant & Identity Analytics Engineer — **May 2022 – Present**

- **Designed and deployed advanced anomaly detection systems** utilizing machine learning algorithms with 95% accuracy to identify anomalous behavioral patterns, credential stuffing, and account takeover vectors across high-volume transaction logs.
- **Engineered scalable big data processing pipelines** using Azure Data Factory, Azure Synapse, and PySpark to ingest, cleanse, and process massive authentication logs and behavioral datasets for real-time identity fraud analytics.
- **Developed predictive machine learning models with XGBoost and Scikit-Learn** to evaluate behavioral biometrics and customer risk profiles, improving operational reliability by 20% and strengthening identity verification frameworks.
- **Automated data quality audits and operational firewalls** utilizing advanced SQL stored procedures and Python regex classification to reduce operational turnaround by over 80% and ensure strict adherence to identity governance metrics.
- **Established executive monitoring dashboards and anomaly tracking metrics** using Power BI (DAX) and Tableau to monitor false-positive rates, model performance, and fraud capture rates, maintaining a 99.9% data availability SLA.

Machine Learning & Data Engineering Lead — **May 2022 – Present**

- **Built high-performance dimensional data models** incorporating Star/Snowflake schemas and Slowly Changing Dimensions (SCD Type 1/2) for secure identity lifecycle management repositories.
- **Implemented automated conversational AI and QA testing frameworks** utilizing prompt engineering and KIRO IDE to accelerate development delivery cycles by 40% while preserving strict security protocols.
- **Optimized cloud data architecture across Azure and AWS** leveraging Snowflake, Redshift, and S3 to manage large-scale transactional datasets and ensure rapid query performance for risk analytics.

Data Science & Predictive Analytics Specialist — **May 2022 – Present**

- **Applied advanced time-series forecasting models** including SARIMA to optimize operational capacity planning and resource allocation, reducing response times by 20%.
- **Conducted root cause analysis and forensic investigations** on production data anomalies, authoring comprehensive runbooks and maintaining continuous system resilience.
- **Collaborated with cross-functional technical stakeholders** to translate complex identity and access management requirements into robust, scalable software engineering solutions.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture — Pune, India

Test Engineering Analyst — **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.
- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.